

Strategy creation using Bollinger Bands

```
In [3]: # Importing Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import yfinance as yf
import datetime
import quantstats as qs
import warnings
warnings.filterwarnings("ignore")
```

```
In [23]: ticker = "ETH-USD"
end = datetime.date(2026, 9 , 29)
start = end - pd.Timedelta(days = 365*1)
```

```
In [27]: class Bollingerbands :

    def __init__(self , ticker , start , end, std):
        self.ticker = ticker
        self.start = start
        self.end = end
        self.std = std
        self.fetch_data()
        self.indicator()
        self.signal()
        self.position()
        self.returns()

    def fetch_data(self):

        self.data = yf.download(self.ticker , self.start , self.end)
        self.data = self.data.droplevel( 1 , axis = 1)

    def indicator(self):

        self.data["20sma"] = self.data["Close"].rolling(window = 20).mean()
        self.data["upper_band"] = self.data["20sma"] + (self.std * self.data["Close"].rolling(window = 20).std())
        self.data["lower_band"] = self.data["20sma"] - (self.std * self.data["Close"].rolling(window = 20).std())

    def signal(self):

        self.data["signal"] = np.where((self.data["Close"] >= self.data["upper_band"]) &
                                       (self.data["Close"].shift(1) < self.data["upper_band"].shift(1)), -1, 0)

        self.data["signal"] = np.where((self.data["Close"] <= self.data["lower_band"])&
                                       (self.data["Close"].shift(1) > self.data["lower_band"].shift(1)), 1, self.data["signal"])

    def position(self):

        self.data["position"] = self.data["signal"].replace(to_replace=0 , method = "ffill")

    def returns(self):

        self.data["bnh"] = np.log(self.data["Close"] / self.data["Close"].shift(1))

        self.data["strategy_returns"] = self.data["bnh"] * self.data["position"].shift(1)
        print("The strategy returns is :", np.round(self.data["strategy_returns"].cumsum()[-1] , 2) )

        return self.data["strategy_returns"].cumsum()[-1]

    def analysis(self):

        self.data[["20sma" , "upper_band" , "lower_band", "position" , "Close"]].plot(figsize = (15,9), secondary_y = "position" , grid = True)
        self.data[["bnh" , "strategy_returns"]].cumsum().plot(figsize = (15,9) , grid = True , linewidth = 0.75)

        qs.reports.basic(self.data["strategy_returns"])
```

```
In [29]: a = Bollingerbands(ticker , start , end, 1.5)
```

[*****100%*****] 1 of 1 completed
The strategy returns is : -0.57

```
In [35]: standard_deviation = [0.5, 0.75 , 1 , 1.25 , 1.50 , 1.75 , 2 , 2.25 ,2.5 ,3]
stock_list = [ "AAPL",
               "TSLA",
               "RELIANCE.NS",
               "INFY.NS",
               "SPY",
               "BTC-USD",
               "ETH-USD",
               "GC=F",
               "CL=F",
               "EURUSD=X"]
std = []
list1 = []
outcome_std = []

for x in standard_deviation:
    for y in stock_list:
        print("For stock std is :" , x , y)

        b = Bollingerbands(ticker, start, end , x)
        std.append(x)
        list1.append(y)
        outcome_std.append(b.returns())

table = pd.DataFrame({"std" : std , "outcome_std" : outcome_std, "list1" : list1})
table
```

For stock std is : 0.5 AAPL
For stock std is : 0.5 TSLA
For stock std is : 0.5 RELIANCE.NS
For stock std is : 0.5 INFY.NS
For stock std is : 0.5 SPY
For stock std is : 0.5 BTC-USD
For stock std is : 0.5 ETH-USD
For stock std is : 0.5 GC=F
For stock std is : 0.5 CL=F
For stock std is : 0.5 EURUSD=X

[*****100%*****] 1 of 1 completed
[*****100%*****] 1 of 1 completed

The strategy returns is : -0.27
The strategy returns is : -0.27
For stock std is : 0.75 AAPL
For stock std is : 0.75 TSLA
For stock std is : 0.75 RELIANCE.NS
For stock std is : 0.75 INFY.NS
For stock std is : 0.75 SPY
For stock std is : 0.75 BTC-USD
For stock std is : 0.75 ETH-USD
For stock std is : 0.75 GC=F
For stock std is : 0.75 CL=F
For stock std is : 0.75 EURUSD=X
The strategy returns is : -0.44
The strategy returns is : -0.44
For stock std is : 1 AAPL
For stock std is : 1 TSLA
For stock std is : 1 RELIANCE.NS
For stock std is : 1 INFY.NS
For stock std is : 1 SPY
For stock std is : 1 BTC-USD
For stock std is : 1 ETH-USD
For stock std is : 1 GC=F
For stock std is : 1 CL=F
For stock std is : 1 EURUSD=X

[*****100%*****] 1 of 1 completed
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The strategy returns is : -0.47
The strategy returns is : -0.47
For stock std is : 1.25 AAPL
For stock std is : 1.25 TSLA
For stock std is : 1.25 RELIANCE.NS
For stock std is : 1.25 INFY.NS
For stock std is : 1.25 SPY
For stock std is : 1.25 BTC-USD
For stock std is : 1.25 ETH-USD
For stock std is : 1.25 GC=F
For stock std is : 1.25 CL=F
For stock std is : 1.25 EURUSD=X
The strategy returns is : -0.45
The strategy returns is : -0.45
For stock std is : 1.5 AAPL
For stock std is : 1.5 TSLA
For stock std is : 1.5 RELIANCE.NS
For stock std is : 1.5 INFY.NS
For stock std is : 1.5 SPY
For stock std is : 1.5 BTC-USD
For stock std is : 1.5 ETH-USD
For stock std is : 1.5 GC=F
For stock std is : 1.5 CL=F
For stock std is : 1.5 EURUSD=X

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[*****100%*****] 1 of 1 completed

The strategy returns is : -0.57
The strategy returns is : -0.57
For stock std is : 1.75 AAPL
For stock std is : 1.75 TSLA
For stock std is : 1.75 RELIANCE.NS
For stock std is : 1.75 INFY.NS
For stock std is : 1.75 SPY
For stock std is : 1.75 BTC-USD
For stock std is : 1.75 ETH-USD
For stock std is : 1.75 GC=F
For stock std is : 1.75 CL=F
For stock std is : 1.75 EURUSD=X
The strategy returns is : -0.8
The strategy returns is : -0.8
For stock std is : 2 AAPL
For stock std is : 2 TSLA
For stock std is : 2 RELIANCE.NS
For stock std is : 2 INFY.NS
For stock std is : 2 SPY
For stock std is : 2 BTC-USD
For stock std is : 2 ETH-USD
For stock std is : 2 GC=F
For stock std is : 2 CL=F
For stock std is : 2 EURUSD=X

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The strategy returns is : -0.44
The strategy returns is : -0.44
For stock std is : 2.25 AAPL
For stock std is : 2.25 TSLA
For stock std is : 2.25 RELIANCE.NS
For stock std is : 2.25 INFY.NS
For stock std is : 2.25 SPY
For stock std is : 2.25 BTC-USD
For stock std is : 2.25 ETH-USD
For stock std is : 2.25 GC=F
For stock std is : 2.25 CL=F
For stock std is : 2.25 EURUSD=X
The strategy returns is : 0.13
The strategy returns is : 0.13
For stock std is : 2.5 AAPL
For stock std is : 2.5 TSLA
For stock std is : 2.5 RELIANCE.NS
For stock std is : 2.5 INFY.NS
For stock std is : 2.5 SPY
For stock std is : 2.5 BTC-USD
For stock std is : 2.5 ETH-USD
For stock std is : 2.5 GC=F
For stock std is : 2.5 CL=F
For stock std is : 2.5 EURUSD=X

[*****100%*****] 1 of 1 completed
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The strategy returns is : -0.43
The strategy returns is : -0.43
For stock std is : 3 AAPL
For stock std is : 3 TSLA
For stock std is : 3 RELIANCE.NS
For stock std is : 3 INFY.NS
For stock std is : 3 SPY
For stock std is : 3 BTC-USD
For stock std is : 3 ETH-USD
For stock std is : 3 GC=F
For stock std is : 3 CL=F
For stock std is : 3 EURUSD=X
The strategy returns is : -0.46
The strategy returns is : -0.46

Out[35]:

	std	outcome_std	list1
0	0.50	-0.266141	EURUSD=X
1	0.75	-0.443851	EURUSD=X
2	1.00	-0.466443	EURUSD=X
3	1.25	-0.453988	EURUSD=X
4	1.50	-0.569114	EURUSD=X
5	1.75	-0.796616	EURUSD=X
6	2.00	-0.435335	EURUSD=X
7	2.25	0.131199	EURUSD=X
8	2.50	-0.432847	EURUSD=X
9	3.00	-0.456066	EURUSD=X

In [36]:

```
f = table.sort_values(by="outcome_std" , ascending=False)
f
```

Out[36]:

	std	outcome_std	list1
7	2.25	0.131199	EURUSD=X
0	0.50	-0.266141	EURUSD=X
8	2.50	-0.432847	EURUSD=X
6	2.00	-0.435335	EURUSD=X
1	0.75	-0.443851	EURUSD=X
3	1.25	-0.453988	EURUSD=X
9	3.00	-0.456066	EURUSD=X
2	1.00	-0.466443	EURUSD=X
4	1.50	-0.569114	EURUSD=X
5	1.75	-0.796616	EURUSD=X

In []: